NICHOLAS WONG

QUANTITATIVE RISK ANALYST

CONTACT

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Sydney, Australia

QUANTITATIVE SKILLS

PROGRAMMING

- Python
- R
- SQL
- C
- Matlab
- VBA

MATHEMATICS

- Functional Approximation Theory
- Stochastic Processes and Analysis
- Statistical Inference and Modelling
- Machine (Deep) Learning
- Optimisation and Approximation
- Functional Analysis
- Time Series Analysis

LEADERSHIP

IDMC SYDNEY

- Operations Director (2023 – 2025)
- Young Adults Leader (2023 – 2025)
- Productions Director (2019 – 2022)

ACTUARIAL SOCIETY OF UNSW (ASOC)

- Events Portfolio (2022)
- Careers Portfolio (2021)
- Peer Mentor (2021 and 2022)
- Authored ASOC Careers Guide (2021)

INTERESTS

Poker, Chess, Volleyball (Captain), Piano, Audio Engineering, Philosophy

EDUCATION

UNIVERSITY OF NEW SOUTH WALES — 2021 - 2025

Actuarial and Mathematics (Honors) | First Class Honours | WAM (Overall): 83

- UG Courses: Computational Math for Finance (92), Statistical Inference (96), Bayesian Methods (94)
- PG Courses: Discrete Time Financial Modelling, Stochastic Analysis, Functional Analysis, Measure Theory

RESEARCH

- Honours research on approximation theory for operator learning methods (Fourier Neural Operators) for PDE surrogate modelling. Derived approximation bounds for groundwater flow PDEs and supported the theory with computational experiments; paper to be submitted to *NeurIPS 2026* as first author
- Co-authored a study of digital skin cancer screening service, contributing to study design, hypothesis formulation, statistical testing, and analysis of triage, follow-up and biopsy outcomes; manuscript submitted to the *Australasian Journal of Dermatology* as second author

AWARDS / COMPETITIONS

- 1st in UNSW's Bayesian Inference (2024)
- Dean's List of Excellence in Academic Performance (2020)
- 1st in SIG Poker Night (2021 and 2022)
- IMC Poker Night Finalist (2024)
- 1st in HSC Mathematics Extension 1/2 (2019)

PROFESSIONAL EXPERIENCE

MACQUARIE GROUP — 2026

Risk Management Group – Market Risk

- Associate within the VaR / Quant Models teams, enhancing Monte Carlo models for daily VaR production across commodity and equity derivatives portfolios with nonlinear options exposure
- Applied options theory through analysis of delta, gamma, vega sensitivities, and volatility surfaces, improving the robustness of risk models through backtesting and general data quality management.

LYRA TECHNOLOGIES — 2025

Machine Learning Researcher Internship

- Built a ML pipeline for a road fatalities predictive model by designing XGBoost and Neural Network architectures, achieving 91% AUC in binary classification to support data-driven road safety initiatives
- Developed a Python-based architecture using novel techniques such as SMOTE for a 1.7% class imbalance; PCA for dimensionality reduction and Bayesian Optimisation for hyperparameter tuning

QUANTITATIVE PROJECTS

STATISTICAL ARBITRAGE CRYPTO

- Engineered a cross-sectional momentum, reversal and breakout strategy for cryptocurrency trading with a 2.34 Sharpe (96% annual return) and 27% Max Drawdown, using a lag-based signal construction
- Backtested strategies in Python over a 5 year period using daily data with 7bp trading costs; implemented smoothing techniques, cross-sectional ranking and statistical robustness tests to refine trading signals

FETCH PET INSURANCE PRICING

- Designed a pet insurance pricing model within a small team, leveraging GAMs and XGBoost for technical pricing and data analysis, achieving 10% MAPE enhancing pricing accuracy and strategic decision-making
- Integrated diverse data sources including shapefiles, socioeconomic data, and geographic data in R to refine feature engineering and improve model performance

MARKOWITZ PORTFOLIO THEORY FOR EQUITIES

- Implemented modern portfolio theory and CAPM to construct tangency portfolios achieving 3+ Sharpe with 0.08 volatility using multivariate optimization on financial data from TSE and Vietnamese equities.
- Applied VBA and optimization techniques in Excel to detect pricing inefficiencies, akin to identifying potential arbitrage opportunities in financial markets.